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Databricks in Action: From Unity Catalog to Business Insights with SQL, Genie & AI/BI Dashboards

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As data professionals, our ultimate goal is not just storing data — it is turning raw numbers into actionable business intelligence.

For my latest hands-on project at **Innomatics Research Labs**, I stepped into the shoes of a Data Analyst for a retail company. The challenge? Build an end-to-end retail sales analytics solution entirely within **Databricks**, leveraging Unity Catalog, Databricks SQL, AI-assisted querying with Genie, and interactive AI/BI Dashboards.

Here is a step-by-step walkthrough of how I built this solution and the insights I uncovered.

1. Introduction: The Business Challenge

Imagine a retail organization managing sales across diverse products, regions, sales representatives, and sales channels. Management needs answers to critical strategic questions:

- What is the overall scale of our recorded sales revenue?
- Which products and regions are driving our performance?
- How do sales trend over time, and which categories contribute the most?

- Can AI-assisted tools like Databricks Genie speed up exploratory analysis while maintaining strict accuracy?

To solve this, I built a fully integrated data pipeline and analytics environment within Databricks.

2. Dataset Overview

The analysis was performed using the **Retail Sales Dataset**, which captures comprehensive transaction records:

- **Dimensions & Metrics:** `Product_ID`, `Sale_Date`, `Sales_Rep`, `Region`, `Sales_Amount`, `Quantity_Sold`, `Product_Category`, `Unit_Cost`, `Unit_Price`, `Customer_Type`, `Discount`, `Payment_Method`, and `Sales_Channel`.
- **Key Rule:** `Sales_Amount` was utilized directly as recorded for revenue analysis to ensure consistency with source tracking.

3. Unity Catalog Setup: Organizing Data

Before diving into analysis, data governance and organization are paramount. In Databricks, data assets are structured hierarchically using **Unity Catalog**:



Plaintext

```
Innomatics (Catalog) → default (Schema) → sales_dataset (Table)
```

Why Organize with Unity Catalog?

- **Catalog:** Acts as the highest-level container, securing and governing all data assets.
- **Schema (Database):** Groups related tables together under a catalog, keeping projects organized.

- **Table:** The structured columnar storage where our retail data resides, ready for SQL querying.

4. Databricks SQL Analysis: Answering Core Business Questions

To address management's inquiries, I wrote and executed five core analytical queries in Databricks SQL.

Task 1: Overall Sales Performance

- **SQL Query:**

```
SELECT
    SUM(Sales_Amount) AS Total_Revenue,
    SUM(Quantity_Sold) AS Total_Units_Sold,
    COUNT(*) AS Total_Transactions,
    COUNT(DISTINCT Product_ID) AS Distinct_Products
FROM sales_dataset;
```

- **Business Insight:** Gives a high-level snapshot of the business scale, total transaction volume, and product catalog breadth.

Task 2: Top Five Products

- **SQL Query:**

```
SELECT
    Product_ID,
    SUM(Sales_Amount) AS Total_Revenue,
    SUM(Quantity_Sold) AS Total_Units_Sold
FROM sales_dataset
GROUP BY Product_ID
ORDER BY Total_Revenue DESC
LIMIT 5;
```

- **Business Insight:** Highlights our top revenue-generating products, helping guide inventory stocking and promotional focus.

Task 3: Regional Performance

- **SQL Query:**

```
SELECT
    Region,
    SUM(Sales_Amount) AS Total_Revenue,
    SUM(Quantity_Sold) AS Total_Units_Sold
FROM sales_dataset
GROUP BY Region
ORDER BY Total_Revenue DESC;
```

- **Business Insight:** Identifies top-performing geographic markets and flags regions requiring targeted marketing or operational investigation.

Task 4: Monthly Sales Trend

- SQL Query:

```
SELECT
    DATE_FORMAT(CAST(Sale_Date AS DATE), 'yyyy-MM') AS Sales_Month,
    SUM(Sales_Amount) AS Total_Revenue
FROM sales_dataset
GROUP BY Sales_Month
ORDER BY Sales_Month ASC;
```

- **Business Insight:** Uncovers seasonal fluctuations and growth trajectories over time.

Task 5: Product Category Contribution

- SQL Query:

```
SELECT
    Product_Category,
    SUM(Sales_Amount) AS Category_Revenue,
    ROUND((SUM(Sales_Amount) / (SELECT SUM(Sales_Amount) FROM sales_dataset)) *
FROM sales_dataset
GROUP BY Product_Category
ORDER BY Category_Revenue DESC;
```

- **Business Insight:** Quantifies exact revenue shares per product category to evaluate portfolio balance.

5. AI-Assisted Business Analysis with Databricks Genie

To explore natural language querying, I used **Databricks Genie** to ask questions

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underlying SQL, every AI-generated query was rigorously cross-verified against my manual SQL results to ensure data accuracy and prevent hallucinations.

6. Databricks AI/BI Dashboard

I transformed the SQL insights into an interactive executive dashboard featuring:

- **KPI Cards:** Total Sales Revenue, Total Quantity Sold, Distinct Products, and Total Transactions.
- **Visualizations:** Bar charts for Regional Performance, Top Products, Monthly Trends, and Category Contributions.
- **Interactive Filters:** Multi-dimensional filtering by `Region`, `Product_Category`, and `Sales_Channel` for dynamic exploration.

7. Conclusion & Recommendations

This end-to-end Databricks project demonstrated how unified data governance (Unity Catalog), robust SQL querying, AI assistance (Genie), and interactive dashboards combine to form a modern analytics powerhouse.

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